

Retail Store Sales — Data Cleaning & Dashboard Validation Report

Verification of the Python (pandas) cleaning pipeline and 3-page Power BI dashboard against the raw dataset

1. Project Overview

The project takes a raw retail store sales dataset and cleans it using Python (pandas): filling/removing null values, removing duplicates, and adding calculated columns (Discount Amount, Discount Percentage, Net/Total Sales). The cleaned data was then loaded into two additional sheets — a Category Summary and a Customer Summary — and visualized across a 3-page Power BI dashboard (an overview page plus two category-detail pages showing item-level tables). This report independently re-checks the raw data, the cleaned output, and all three dashboard pages against each other.

2. Raw Data vs. Cleaned Data

The raw CSV (retail_store_sales.csv) contains 12,575 rows, with missing values in Item (1,213), Price Per Unit (609), Quantity (604), Total Spent (604), and Discount Applied (4,199 — expected, since not every transaction has a discount).

The cleaning script drops rows missing Quantity or Total Spent. 12,575 minus 604 = **11,971 rows**, matching the cleaned 'store_details' sheet exactly. The cleaned sheet has **zero nulls and zero duplicate rows**. Result: verified correct.

3. Calculated Column Validation

Derived column	Formula	Match rate
Sales_Amount	Price Per Unit x Quantity	100%
Net_Sales	Sales_Amount - Discount_Amount	100%
Discount_Percentage	Discount_Amount / Sales_Amount	100% (0% or 10% only)

All three formulas are consistent for every one of the 11,971 rows. Result: verified correct.

4. Power BI Dashboard — Page 1 (Overview KPIs)

Metric	Recomputed from data	Shown on dashboard	Match
Sum of Net Sales	1,550,329.95	1.55M	Yes
Sum of Net Profit	320,977.70	320.98K	Yes
Unique customers	25	25	Yes
Total transactions	11,971	12K	Yes

All 4 overview KPIs tie out exactly to the cleaned data. Result: verified correct.



Page 1 — Overview dashboard (KPI cards, category charts, location map)

5. Power BI Dashboard — Pages 2 & 3, Category Totals

The 'Total' row at the bottom of each category block on pages 2 and 3 was checked against the cleaned workbook. Total Spent and Net Profit come from the 'category summary' sheet; Discount Amount was cross-checked against the real per-transaction discounts summed from store_details:

Category	Total Spent	Discount Amt	Net Profit	Match
Beverages	163,357.20	6,964.75	40,839.30	Yes
Food	160,726.40	6,406.95	40,181.60	Yes
Milk Products	143,201.20	6,005.25	35,800.30	Yes
Butchers	170,641.60	7,492.60	42,660.40	Yes
Electric Household Ess.	168,122.00	7,042.70	42,030.50	Yes
Computers & Accessories	156,127.60	6,495.80	39,031.90	Yes
Patisserie	156,513.20	7,176.05	39,128.30	Yes
Furniture	165,221.60	6,938.95	41,305.40	Yes

All 8 category Total rows tie out exactly, including Discount Amount, which in this latest version of the dashboard now correctly totals the real per-transaction discount for each category (an earlier version of this table showed much lower, incorrect discount totals for 5 of the 8 categories — that has since been corrected). Result: verified correct.

lipboard

Data					Queries	Insert	Calculations		
Category	Item Name	Sum of Total_spent	Sum of Discount_Amount	Sum of Net_profit	Category	Item Name	Sum of Total_spent	Sum of Discount_Amount	Sum of Net_profit
Beverages	Espresso	18,400.80	6,964.75	4,600.20	Food	Basmati Rice	660.40	6,406.95	165.10
Beverages	Cola	13,050.80	6,964.75	3,262.70	Food	Biscuits	10,752.00	6,406.95	2,688.00
Beverages	Strawberry Smoothie	12,462.00	6,964.75	3,115.50	Food	Bread	4,050.00	6,406.95	1,012.50
Beverages	Chai Latte	11,152.40	6,964.75	2,788.10	Food	Canned Beans	2,412.80	6,406.95	603.20
Beverages	Apple Juice	11,020.00	6,964.75	2,755.00	Food	Canned Corn	2,376.00	6,406.95	594.00
Beverages	Green Tea	9,546.00	6,964.75	2,386.50	Food	Cereal	1,748.40	6,406.95	437.10
Beverages	Fresh Orange Juice	9,438.00	6,964.75	2,359.50	Food	Cooking Oil	9,697.60	6,406.95	2,424.40
Beverages	Mango Juice	8,678.40	6,964.75	2,169.60	Food	Corn Flakes	4,410.40	6,406.95	1,102.60
Beverages	Milkshake	8,288.00	6,964.75	2,072.00	Food	Instant Soup	2,401.60	6,406.95	600.40
Beverages	Mineral Water	7,592.00	6,964.75	1,898.00	Food	Jam	11,740.40	6,406.95	2,935.10
Beverages	Sparkling Water	6,536.00	6,964.75	1,634.00	Food	Mayonnaise	9,597.60	6,406.95	2,399.40
Total		1,63,357.20	6,964.75	40,839.30	Total		1,60,726.40	6,406.95	40,181.60

Category	Item Name	Sum of Total_spent	Sum of Discount_Amount	Sum of Net_profit	Category	Item Name	Sum of Total_spent	Sum of Discount_Amount	Sum of Net_profit
Milk Products	Lassi	392.00	6,005.25	98.00	Furniture	Bar Stool	11,564.80	6,938.95	2,891.20
Milk Products	Butter	707.20	6,005.25	176.80	Furniture	Bean bag chair	7,544.00	6,938.95	1,886.00
Milk Products	Salted Butter	1,079.20	6,005.25	269.80	Furniture	Bed Frame	4,732.80	6,938.95	1,183.20
Milk Products	Full Cream Milk	1,456.00	6,005.25	364.00	Furniture	Bedside Table	6,807.60	6,938.95	1,701.90
Milk Products	Fresh Cream	1,599.60	6,005.25	399.90	Furniture	Bookshelf	5,450.00	6,938.95	1,362.50
Milk Products	Plain Yogurt	1,628.00	6,005.25	407.00	Furniture	Cabinet	3,336.80	6,938.95	834.20
Milk Products	Condensed Milk	1,646.40	6,005.25	411.60	Furniture	Coffee Table	1,702.40	6,938.95	425.60
Milk Products	Greek Yogurt	2,006.40	6,005.25	501.60	Total		1,65,221.60	6,938.95	41,305.40
Milk Products	Fresh Milk	2,208.00	6,005.25	552.00					
Total		1,43,201.20	6,005.25	35,800.30					

Category detail — Beverages, Food, Milk Products, Furniture

Category	Item Name	Sum of Total_spent	Sum of Discount_Amount	Sum of Net_profit	Category	Item Name	Sum of Total_spent	Sum of Discount_Amount	Sum of Net_profit
Butchers	Chicken Breast	20,631.20	7,492.60	5,157.80	Electric household essentials	Electric Kettle	22,304.00	7,042.70	5,576.00
Butchers	Pork Ribs	16,235.20	7,492.60	4,058.80	Electric household essentials	Electric Grill	15,747.20	7,042.70	3,936.80
Butchers	Mutton Biryani Cut	15,291.20	7,492.60	3,822.80	Electric household essentials	Water Heater	13,534.00	7,042.70	3,383.50
Butchers	Beef Roast	15,088.40	7,492.60	3,772.10	Electric household essentials	Vacuum Cleaner	11,814.40	7,042.70	2,953.60
Butchers	Chicken Sausage	10,406.00	7,492.60	2,601.50	Electric household essentials	Ceiling Fan	10,648.80	7,042.70	2,662.20
Butchers	Beef Liver	9,760.00	7,492.60	2,440.00	Electric household essentials	Food Processor	10,395.20	7,042.70	2,598.80
Butchers	Turkey Breast	9,256.00	7,492.60	2,314.00	Electric household essentials	Iron	9,476.00	7,042.70	2,369.00
Butchers	Chicken Drumsticks	8,932.00	7,492.60	2,233.00	Electric household essentials	Electric Shaver	8,985.60	7,042.70	2,246.40
Butchers	Mutton Mince	8,926.80	7,492.60	2,231.70	Electric household essentials	Juicer	8,176.00	7,042.70	2,044.00
Butchers	Mixed Meat Balls	8,232.00	7,492.60	2,133.80	Electric household essentials	Hand Mixer	7,241.60	7,042.70	1,810.40
Total		1,70,641.60	7,492.60	42,660.40	Total		1,68,122.00	7,042.70	42,030.50
Category	Item Name	Sum of Total_spent	Sum of Discount_Amount	Sum of Net_profit	Category	Item Name	Sum of Total_spent	Sum of Discount_Amount	Sum of Net_profit
Computers and electric accessories	Computer Monitor	1,081.60	6,495.80	270.40	Computers and electric accessories	Desktop Computer	2,168.40	6,495.80	542.10
Computers and electric accessories	Desktop Computer	2,168.40	6,495.80	542.10	Computers and electric accessories	Ethernet Cable	6,366.40	6,495.80	1,591.60
Computers and electric accessories	Ethernet Cable	6,366.40	6,495.80	1,591.60	Computers and electric accessories	External Hard Drive	9,588.80	6,495.80	2,397.20
Computers and electric accessories	External Hard Drive	9,588.80	6,495.80	2,397.20	Computers and electric accessories	HDMI Cable	10,096.40	6,495.80	2,524.10
Computers and electric accessories	HDMI Cable	10,096.40	6,495.80	2,524.10	Computers and electric accessories	Headphones	5,644.00	6,495.80	1,411.00
Computers and electric accessories	Headphones	5,644.00	6,495.80	1,411.00	Computers and electric accessories	Keyboard	646.00	6,495.80	161.50
Computers and electric accessories	Keyboard	646.00	6,495.80	161.50	Computers and electric accessories	Laptop	8,626.40	6,495.80	2,156.60
Computers and electric accessories	Laptop	8,626.40	6,495.80	2,156.60	Computers and electric accessories	Laptop Charger	6,403.20	6,495.80	1,600.80
Computers and electric accessories	Laptop Charger	6,403.20	6,495.80	1,600.80	Computers and electric accessories	Laptop Stand	8,555.60	6,495.80	2,138.90
Computers and electric accessories	Laptop Stand	8,555.60	6,495.80	2,138.90	Total		1,56,127.60	6,495.80	39,031.90
Total		1,56,127.60	6,495.80	39,031.90					
Category	Item Name	Sum of Total_spent	Sum of Discount_Amount	Sum of Net_profit	Category	Item Name	Sum of Total_spent	Sum of Discount_Amount	Sum of Net_profit
Patisserie	Croissant	15,350.40	7,176.05	3,837.60	Patisserie	Chocolate Tart	15,048.00	7,176.05	3,762.00
Patisserie	Chocolate Tart	15,048.00	7,176.05	3,762.00	Patisserie	Strawberry Tart	13,802.00	7,176.05	3,450.50
Patisserie	Strawberry Tart	13,802.00	7,176.05	3,450.50	Patisserie	Madeleine	11,716.00	7,176.05	2,929.00
Patisserie	Madeleine	11,716.00	7,176.05	2,929.00	Patisserie	Vanilla Slice	11,534.00	7,176.05	2,883.50
Patisserie	Vanilla Slice	11,534.00	7,176.05	2,883.50	Patisserie	Caramel Cake	9,614.80	7,176.05	2,402.30
Patisserie	Caramel Cake	9,614.80	7,176.05	2,402.30	Total		1,56,513.20	7,176.05	39,128.30
Total		1,56,513.20	7,176.05	39,128.30					

Category detail — Butchers, Electric Household Essentials, Computers & Accessories, Patisserie

6. Critical Finding — Item-Level Rows Do Not Match the Data

While the category Total rows are accurate, the individual item rows above them (e.g. Espresso, Chicken Breast, Computer Monitor) still do not reconcile with the store_details sheet. Testing multiple items found a consistent, exact pattern:

- "Total_spent" shown per item = exactly 80% of that item's real Sales_Amount in the cleaned data.
- "Net_profit" shown per item = exactly 25% of that (already understated) Total_spent figure — a flat assumed margin, not the real Net_Sales.
- "Discount_Amount" shown per item is identical for every item within a category (e.g. every single Butchers item shows 7,492.60) — this is the category's total discount repeated on each row, not that item's own discount. This category-level figure is now correct (see Section 5), but it is still not broken down by item.

Example: Chicken Breast's real Sales_Amount is 25,789.00. The dashboard shows 20,631.20 ($25,789.00 \times 0.80$) and a Net_profit of 5,157.80 ($20,631.20 \times 0.25$) — neither figure is the item's actual sales or profit. The same pattern holds for newer items such as Computer Monitor (real Sales_Amount 1,352.00 vs. dashboard's 1,081.60) and Lassi (real 490.00 vs. dashboard's 392.00).

This means the item-level table visual on pages 2 and 3 is built on a different measure than the Category Summary sheet (likely a flat-percentage DAX measure instead of referencing the real Discount_Amount / Net_Sales columns per transaction). This should be fixed before submission — the accurate item-level figures, recomputed directly from store_details, are given below for reference.

7. Item-Level Totals — Recomputed From Cleaned Data (Corrected)

The tables below are the accurate per-item Total Spent, Discount Amount, and Net Profit, computed directly from the store_details sheet, for use in place of the dashboard's item-level rows.

Beverages

Item	Total Spent	Discount Amt	Net Profit
Espresso	23,001.00	840.50	22,160.50
Cola	16,313.50	651.75	15,661.75
Strawberry Smoothie	15,577.50	438.85	15,138.65
Chai Latte	13,940.50	431.20	13,509.30
Apple Juice	13,775.00	452.40	13,322.60
Green Tea	11,932.50	346.15	11,586.35
Fresh Orange Juice	11,797.50	297.00	11,500.50
Mango Juice	10,848.00	438.40	10,409.60
Milkshake	10,360.00	367.50	9,992.50
Mineral Water	9,490.00	324.85	9,165.15
Sparkling Water	8,170.00	338.20	7,831.80
Herbal Tea	7,644.00	239.20	7,404.80
Lemonade	7,625.00	323.30	7,301.70
Cold Brew	6,880.00	242.00	6,638.00
Hot Chocolate	6,479.00	289.85	6,189.15
Macchiato	5,866.00	215.60	5,650.40
Americano	4,394.00	133.90	4,260.10
Iced Latte	3,910.00	119.00	3,791.00
Iced Coffee	3,848.00	116.55	3,731.45
Masala Tea	3,649.00	131.20	3,517.80
Café Latte	2,831.00	85.50	2,745.50
Cappuccino	2,392.00	55.20	2,336.80
Flat White	1,875.00	46.25	1,828.75
Black Tea	828.00	18.40	809.60
Mocha	770.00	22.00	748.00
Total	204,196.50	6,964.75	197,231.75

Butchers

Item	Total Spent	Discount Amt	Net Profit
Chicken Breast	25,789.00	926.60	24,862.40
Pork Ribs	20,294.00	912.50	19,381.50
Mutton Biryani Cut	19,114.00	763.80	18,350.20
Beef Roast	18,860.50	318.25	18,542.25
Chicken Sausage	13,007.50	616.00	12,391.50
Beef Liver	12,200.00	237.90	11,962.10
Turkey Breast	11,570.00	455.00	11,115.00
Chicken Drumsticks	11,165.00	437.50	10,727.50
Mutton Mince	11,158.50	423.55	10,734.95
Mixed Meat Pack	10,665.00	316.00	10,349.00
Pork Chops	10,281.00	400.20	9,880.80
Lamb Shank	7,104.00	240.00	6,864.00
Chicken Liver	5,597.00	165.30	5,431.70
Beef Steak	4,737.50	170.00	4,567.50
Mutton Curry Cut	4,500.00	164.00	4,336.00
Crab	4,387.00	118.90	4,268.10
Beef Mince	3,920.00	102.20	3,817.80
Beef Ribs	3,596.00	80.60	3,515.40
Whole Chicken	3,464.00	116.00	3,348.00
Minced Chicken	3,014.00	123.20	2,890.80
Pork Sausage	2,205.00	112.70	2,092.30
Lamb Leg	1,979.50	98.05	1,881.45
Chicken Thigh	1,937.00	85.80	1,851.20
Chicken Wings	1,719.50	54.15	1,665.35
Lamb Chops	1,037.00	54.40	982.60
Total	213,302.00	7,492.60	205,809.40

Computers and electric accessories

Item	Total Spent	Discount Amt	Net Profit
Power Bank	18,496.00	518.40	17,977.60
USB Hub	14,237.50	589.60	13,647.90
HDMI Cable	12,620.50	387.00	12,233.50

Item	Total Spent	Discount Amt	Net Profit
Mouse Pad	12,495.00	420.00	12,075.00
External Hard Drive	11,986.00	358.80	11,627.20
Laptop	10,783.00	282.90	10,500.10
Laptop Stand	10,694.50	379.60	10,314.90
SSD Drive	9,515.00	266.75	9,248.25
USB Flash Drive	9,089.50	347.90	8,741.60
Phone Charger	9,089.00	353.80	8,735.20
Printer	8,398.00	277.40	8,120.60
Laptop Charger	8,004.00	287.10	7,716.90
Ethernet Cable	7,958.00	276.00	7,682.00
USB Cable	7,920.00	290.00	7,630.00
USB Headset	7,751.50	329.30	7,422.20
Headphones	7,055.00	195.50	6,859.50
Wireless Keyboard	6,347.00	199.10	6,147.90
Wireless Mouse	3,962.00	79.80	3,882.20
Webcam	3,844.00	167.40	3,676.60
Smart Plug	3,567.00	86.10	3,480.90
Mouse	3,475.00	121.25	3,353.75
Router	3,002.00	106.65	2,895.35
Desktop Computer	2,710.50	92.95	2,617.55
Computer Monitor	1,352.00	46.40	1,305.60
Keyboard	807.50	36.10	771.40
Total	195,159.50	6,495.80	188,663.70

Electric household essentials

Item	Total Spent	Discount Amt	Net Profit
Electric Kettle	27,880.00	1,016.80	26,863.20
Electric Grill	19,684.00	577.60	19,106.40
Water Heater	16,917.50	680.05	16,237.45
Vacuum Cleaner	14,768.00	436.80	14,331.20
Ceiling Fan	13,311.00	455.30	12,855.70
Food Processor	12,994.00	609.55	12,384.45

Item	Total Spent	Discount Amt	Net Profit
Iron	11,845.00	340.40	11,504.60
Electric Shaver	11,232.00	278.40	10,953.60
Juicer	10,220.00	357.00	9,863.00
Hand Mixer	9,052.00	306.90	8,745.10
Induction Cooktop	8,750.50	331.10	8,419.40
Coffee Maker	6,035.00	173.40	5,861.60
Room Heater	5,978.00	134.20	5,843.80
Table Fan	5,472.50	156.75	5,315.75
Air Fryer	5,440.00	168.00	5,272.00
Wall Shelf	5,043.00	213.20	4,829.80
Electric Cooker	4,906.00	144.10	4,761.90
Blender	4,425.00	131.25	4,293.75
Sandwich Maker	3,977.50	98.05	3,879.45
Steam Iron	3,239.00	19.75	3,219.25
Rice Cooker	3,040.00	107.35	2,932.65
Hair Dryer	2,131.50	151.90	1,979.60
Mixer Grinder	1,806.00	79.80	1,726.20
Toaster	1,573.00	63.05	1,509.95
Microwave Oven	432.00	12.00	420.00
Total	210,152.50	7,042.70	203,109.80

Food

Item	Total Spent	Discount Amt	Net Profit
White Rice	22,509.00	758.50	21,750.50
Sugar	18,693.00	646.55	18,046.45
Jam	14,675.50	455.70	14,219.80
Peanut Butter	14,122.00	466.90	13,655.10
Biscuits	13,440.00	651.00	12,789.00
Olive Oil	13,084.50	326.35	12,758.15
Cooking Oil	12,122.00	400.20	11,721.80
Potato Chips	12,008.50	357.70	11,650.80
Mayonnaise	11,997.00	335.40	11,661.60

Item	Total Spent	Discount Amt	Net Profit
Tomato Sauce	9,060.00	286.00	8,774.00
Spices	8,058.00	142.20	7,915.80
Oats	7,208.00	253.30	6,954.70
Noodles	6,006.00	149.60	5,856.40
Corn Flakes	5,513.00	227.55	5,285.45
Bread	5,062.50	142.50	4,920.00
Prawn curry	4,141.00	135.30	4,005.70
Pasta	3,667.00	121.60	3,545.40
Salt	3,040.00	83.20	2,956.80
Canned Beans	3,016.00	119.60	2,896.40
Instant Soup	3,002.00	83.60	2,918.40
Canned Corn	2,970.00	85.25	2,884.75
Wheat Flour	2,696.00	64.00	2,632.00
Cereal	2,185.50	65.10	2,120.40
Whole Wheat Bread	1,806.00	33.60	1,772.40
Basmati Rice	825.50	16.25	809.25
Total	200,908.00	6,406.95	194,501.05

Furniture

Item	Total Spent	Discount Amt	Net Profit
Dining Chair	30,053.00	1,209.50	28,843.50
Storage Cabinet	21,527.50	628.05	20,899.45
Side Table	14,740.00	412.05	14,327.95
Bar Stool	14,456.00	366.60	14,089.40
Cupboard	14,326.00	288.80	14,037.20
Computer Desk	13,405.00	623.00	12,782.00
TV Stand	11,760.00	328.00	11,432.00
Bean bag chair	9,430.00	356.70	9,073.30
Bedside Table	8,509.50	358.05	8,151.45
Desk	7,837.50	195.25	7,642.25
Folding Chair	7,154.00	274.40	6,879.60
Storage Shelf	7,105.00	248.50	6,856.50

Item	Total Spent	Discount Amt	Net Profit
Bookshelf	6,812.50	193.75	6,618.75
Study Table	6,765.00	183.70	6,581.30
Bed Frame	5,916.00	275.50	5,640.50
Cabinet	4,171.00	172.00	3,999.00
Wardrobe	3,934.00	161.00	3,773.00
Two-Seater Sofa	3,774.00	127.65	3,646.35
Office Chair	3,731.00	135.20	3,595.80
Shoe Rack	3,202.50	115.90	3,086.60
Coffee Table	2,128.00	49.40	2,078.60
Wooden Bench	1,788.50	83.95	1,704.55
Sofa	1,564.00	45.90	1,518.10
Drawer Unit	1,357.00	43.70	1,313.30
Dining Table	1,080.00	62.40	1,017.60
Total	206,527.00	6,938.95	199,588.05

Milk Products

Item	Total Spent	Discount Amt	Net Profit
Chocolate Milk	19,904.00	649.60	19,254.40
Paneer	17,545.00	547.25	16,997.75
Parmesan Cheese	17,062.00	672.60	16,389.40
Cottage Cheese	14,297.00	417.60	13,879.40
Whipping Cream	13,064.00	437.00	12,627.00
Gouda Cheese	11,862.50	375.95	11,486.55
Cream Cheese	11,700.00	376.00	11,324.00
Mango Yogurt	10,507.00	391.05	10,115.95
Sour Cream	8,448.50	277.55	8,170.95
Buttermilk	7,604.50	271.35	7,333.15
Mozzarella Cheese	7,068.00	181.35	6,886.65
Cheddar Cheese	6,594.00	224.00	6,370.00
Strawberry Yogurt	6,112.50	192.50	5,920.00
Low Fat Milk	4,504.00	175.20	4,328.80
Milk Powder	3,640.00	98.80	3,541.20

Item	Total Spent	Discount Amt	Net Profit
Vanilla Yogurt	3,185.00	171.50	3,013.50
Fresh Milk	2,760.00	99.50	2,660.50
Greek Yogurt	2,508.00	68.40	2,439.60
Condensed Milk	2,058.00	120.05	1,937.95
Plain Yogurt	2,035.00	24.20	2,010.80
Fresh Cream	1,999.50	70.95	1,928.55
Full Cream Milk	1,820.00	70.85	1,749.15
Salted Butter	1,313.50	48.10	1,265.40
Butter	884.00	28.90	855.10
Lassi	490.00	15.00	475.00
Total	178,966.00	6,005.25	172,960.75

Patisserie

Item	Total Spent	Discount Amt	Net Profit
Croissant	19,188.00	713.40	18,474.60
Chocolate Tart	18,810.00	513.00	18,297.00
Strawberry Tart	17,252.50	703.50	16,549.00
Madeleine	14,645.00	649.60	13,995.40
Vanilla Slice	14,417.50	762.35	13,655.15
Carrot Cake	12,018.50	408.50	11,610.00
Red Velvet Cake	11,040.00	330.00	10,710.00
Apple Turnover	9,782.00	372.30	9,409.70
Almond Tart	8,736.00	223.60	8,512.40
Pain au Chocolat	8,635.00	264.00	8,371.00
Macaron	8,501.50	328.30	8,173.20
Chocolate Cake	7,015.00	248.40	6,766.60
Fruit Tart	6,417.00	226.30	6,190.70
Lemon Tart	5,472.00	291.20	5,180.80
Cheesecake	5,253.00	181.90	5,071.10
Éclair	4,480.00	133.00	4,347.00
Chocolate Muffin	4,408.00	126.35	4,281.65
Cinnamon Roll	4,087.50	112.50	3,975.00

Item	Total Spent	Discount Amt	Net Profit
Banana Bread	3,965.00	143.35	3,821.65
Donut	3,403.00	110.70	3,292.30
Brownie	2,812.00	142.45	2,669.55
Chocolate Croissant	1,573.00	46.15	1,526.85
Apple Danish	1,551.00	60.50	1,490.50
Custard Danish	1,155.00	45.50	1,109.50
Blueberry Muffin	1,024.00	39.20	984.80
Total	195,641.50	7,176.05	188,465.45

8. Other Minor Issues (recommended fixes before submission)

- store_details sheet has two extra trailing blank/unnamed columns (fully null) — leftover from Excel export. Safe to delete.
- Customer Summary sheet uses generic headers (Column1-Column5) with the real header row sitting as the first row of data.
- Location values are inconsistent for two customers — "Kochi" is written as "kochi" for CUST_16 and CUST_17.

9. Conclusion

The cleaning pipeline, all row-level calculated fields, the overview KPIs, and the category-level totals on the dashboard are all verified accurate. However, the item-level table visual on Pages 2 and 3 uses a different, incorrect calculation (a flat 80%/25% adjustment instead of the real transaction data) and should be rebuilt using the actual Sales_Amount, Discount_Amount, and Net_Sales columns before this is submitted — corrected figures are provided in Section 7.

Report generated by validating the uploaded raw CSV, cleaned Excel workbook, and all three dashboard page screenshots against each other.